

In this article, we introduce the concept of BigchainDB, which is used for database-style decentralised storage.

BigchainDB

BigchainDB allows developers and enterprises to deploy blockchain proof-of-concepts, platforms and applications with a scalable blockchain database. It supports a wide range of industries and use cases, from identity and intellectual property to supply chains, energy, IoT and financial ecosystems — all without sacrificing scale, security or performance. With high throughput, sub-second latency and powerful crypto-condition escrow functionality to automate release-of-assets, BigchainDB looks, acts and feels like a database but has core blockchain characteristics that enterprises want.

As compared to Ethereum, Hyperledger or other blockchain systems, BigchainDB doesn't build a full stack of blockchain technologies, but rather, offers an overlay onto existing database technologies to 'blockchain-ify' them. BigchainDB is designed to merge the best of the database and blockchain worlds — scale and querying from the database side; and decentralisation, immutability and assets from the blockchain side. It starts with an initial open source database, i.e., MongoDB, and adds blockchain characteristics including decentralised control, immutability, and the creation and movement of digital assets. BigchainDB easily accommodates 1 million transactions per second, and explicitly excludes having a virtual machine or other mechanism for running smart contracts.

The following are the three main characteristics of BigchainDB.

- **Decentralised control:** No single entity controls the network. Every node has its own locally-stored list of the public keys of other consortium members — the so-called keyring. There's no centrally-stored or centrally-shared keyring.

- **Immutability:** Data once written cannot be changed or altered. For example, the decimal digits of π are immutable (3.14159...)
- **Transfer of digital assets:** It has the ability to create an asset and transfer it without central control.

A BigchainDB instance consists of a number of nodes, all of which contain parts (but not all) of the complete database. Decentralised control is achieved by this DNS-like federation of nodes, which have voting rights in the validation of blocks. Voting operates on a layer above the actual database and in order to achieve speed, each block of transactions is written before being validated by a quorum of nodes. Nodes vote to validate a transaction and at validation time 'chainify' the block, as each block provides a hash ID of the previous block. Immutability is achieved through a combination of shard replication, disallowing reversions, database backups and the cryptographic signing of all transactions.

BigchainDB technology ensures an efficient decentralised database. It supports millions of writing operations per second, storing PB-level data and delivering a sub-second response time.

Related terminology

To understand the in-depth technical functionalities of BigchainDB, it is important to understand the related terminology.

- **BigchainDB node:** This is a machine or logical machine running the BigchainDB server and other software, and every node is controlled by one person or organisation. A set of BigchainDB nodes can connect to each other to form a cluster. Each node in the cluster runs the same software. A cluster contains one logical RethinkDB datastore. A cluster may have additional machines to do things such as cluster monitoring. There are several kinds of nodes:
1. A dev/test node is one created by a developer working on the

BigchainDB server, e.g., for testing a new or changed code. A dev/test node is typically run on the developer's local machine.

2. A bare-bones node is deployed in the cloud, either as part of a testing cluster or as a starting point before upgrading the node to production-readiness. Our cloud deployment starter templates deploy a bare-bones node, as do our scripts for deploying a testing cluster on AWS.
 3. A production node is part of a federation's BigchainDB cluster. A production node has the most components and requirements.
- **BigchainDB network:** BigchainDB nodes joined together form a BigchainDB network, where every node runs the same software and more nodes are added from time to time. In addition, some machines are especially added to perform monitoring.
 - **BigchainDB consortium:** All the people and organisations that run the nodes in a BigchainDB network are associated with the BigchainDB consortium, which is another organisation, similar to a government structure, for making decisions. If a BigchainDB network is run by a single company, then the consortium is just that company.

Transactions in BigchainDB

The BigchainDB framework makes use of the consensus of the used database cluster. It is also possible for the end users to make their own validation mechanism, to validate the transactions in the blockchain network.

Basically, two types of transactions are available in BigchainDB, i.e., CREATE and TRANSFER. CREATE transactions create an asset, i.e., a collection of data and link it to at least one owner — its public key. In addition, all assets are 'divisible' and can therefore be broken down into different proportions for different owners. In a CREATE transaction, there is no previous owner. So an input in a